Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. What are the two main steps in NER? 1 / 1 point

- Summarization and ranking
- Recognition and classification
- Tokenization and stemming
- Parsing and translation

Nice work

Correct: NER involves first identifying the relevant span of text (recognition) and then assigning the correct entity category (classification). This two-step process ensures both identification and contextual labeling.

2. Scenario: Your model identifies 'IBM Watson Health' as B-ORG, I-ORG, O. What error has likely occurred? 1 / 1 point

- Correct tagging
- Incorrect span termination
- Missing transition probability
- BIO format not used

Nice work

Correct: The model prematurely ended the entity span, possibly due to limitations in training data or a confidence threshold issue.

3. Scenario: A system outputs [O, O, B-LOC, I-LOC, I-LOC O] for the phrase “from the city of Boston today.” Which entity is being labeled? 1 / 1 point

- city
- today
- Boston
- city of Boston

Nice work

Correct: The model marks “city of Boston” using B-LOC and I-LOC, showing it identifies a multi-token location.

4. Your model predicts labels for the sentence "University of Michigan is a great school." You receive the output labels: [B-ORG, I-ORG, O, O, O, O]. Which labeling scheme is being used? 1 / 1 point

- BIO tagging
- Dependency parsing
- POS tagging
- Sentence segmentation

Nice work

Correct. BIO tagging allows token-level labels to denote entity structure by specifying the start (B) and continuation (I) of named entities, while O indicates non-entity tokens.

5. Which of the following features can be used in CRF-based models for NER? 1 / 1 point

- [x] Word suffixes like “-itis”

Nice work

This is one of the correct answers: Morphological patterns (like "-itis" for diseases) offer domain-specific signals.

- [x] Part-of-speech tags

Nice work

This is one of the correct answers. POS tags help distinguish nouns from other word types, aiding entity classification.

- [x] Capitalization of the word

Nice work

This is one of the correct answers. Capitalization patterns often signal proper nouns and are useful cues for named entities.

- [ ] Pretrained transformer embeddings

6. Scenario: You’re designing a model to tag diseases in text. It labels 'Type 2 diabetes' as [B-DISEASE, I-DISEASE]. What does this indicate about the model? 1 / 1 point

- It is ignoring token positions
- It is labeling using POS tags
- It is performing chunking only
- It is applying sequence labeling using BIO format

Nice work

Correct. The model uses BIO encoding to indicate that 'Type' starts the disease name (B-DISEASE), and '2 diabetes' continues it (I-DISEASE), showing proper sequence-aware labeling.

7. A hospital note says "Jane Doe was diagnosed on March 10 with diabetes." Which of the following would typically be extracted as named entities? *(Select all that apply)* 1 / 1 point

- [x] diabetes

Nice work

Correct: “diabetes” is a biomedical entity, frequently labeled in clinical NER tasks.

- [x] Jane Doe

Nice work

Correct: “Jane Doe” refers to a person, which is one of the core entity types in NER.

- [x] March 10

Nice work

Correct: “March 10” is a temporal entity and is commonly extracted as a DATE.

- [ ] diagnosed

8. What is a key assumption (and limitation) of Hidden Markov Models (HMMs)? 1 / 1 point

- Observations depend on the entire input sequence
- Future states influence current observations
- The model uses pre-trained embeddings
- Observations depend only on the current hidden state

Nice work

Correct. HMMs assume that the current observed word or token is generated solely from the current hidden state, without access to previous or future tokens. This simplifies modeling but reduces contextual accuracy.

9. Which component is not part of a Hidden Markov Model? 1 / 1 point

- Emission probabilities
- Transition probabilities
- Initial state probabilities
- Pretrained word embeddings

Nice work

Correct: Word embeddings are vector representations from deep models like BERT or Word2Vec and are not part of traditional HMMs.

10. Which algorithm is commonly used in HMMs to find the most likely label sequence? 1 / 1 point

- Viterbi Algorithm
- Backpropagation
- Naive Bayes
- K-means

Nice work

This is correct: The Viterbi algorithm efficiently computes the most probable path of hidden states given a sequence of observations using dynamic programming.

11. BIO encoding is used to mark where a named entity begins and ends in a sequence. 1 / 1 point

- True
- False

Nice work

Correct. BIO (Begin, Inside, Outside) encoding is a structured labeling scheme that helps the model understand not just what tokens belong to an entity, but also where those entities start and end. This is crucial for span-level labeling.

12. What does Named Entity Recognition (NER) involve? 1 / 1 point

- Generating new text
- Translating one language to another
- Removing stopwords from text
- Recognizing and classifying named entities in text

Nice work

Correct: Named Entity Recognition is a core NLP task that involves identifying specific spans of text that refer to distinct real-world entities (such as “United Nations” or “New York City”) and labeling them with predefined categories like PERSON, LOCATION, ORGANIZATION, DATE, etc.